



Spoken Language Processing - Summer 2026

Course Project

Submission Deadline: Sat. 19/9/2026 on ITC

Working in groups:

As you know, this course requires a comprehensive term project that will constitute 15% of your course grade. Since most of these projects require an intensive workload, you have to work on this project in groups of **three** (four is not allowed!).

Project report

Final project reports are due on Sat. 19/9/2026. They should be submitted through Moodle.

Final project report should be 2-4 pages long in the IEEE transaction letter format (including appendices, figures, references, and everything else you choose to submit). The following is a suggested structure for the final report:

1. Title, Author(s), Affiliation

2. Abstract: The abstract should provide a concise summary of the entire project (typically 150–250 words). It should briefly describe:

The project objective and motivation.

The methodology or approach used.

The main implementation and experiments performed.

The key results and findings.

The overall outcome of the project.

The abstract should be self-contained and should allow the reader to understand the essence of the project without reading the full report.

3. Introduction: this section introduces your problem, and the overall plan for approaching your problem

4. **Background/Related Work:** This section discusses relevant literature for your project

5. **Methodology** (system description): This section details the framework of your project. Be specific, which means you might want to include equations, figures, plots, etc

6. **Experiments and Results:** This section begins with what kind of experiments you're doing, what kind of dataset(s) you're using, and what is the way you measure or evaluate your results. It then shows in details the results of your experiments. By details, we mean both quantitative evaluations (show numbers, figures, tables, etc) as well as qualitative results (show images, example results, etc).

7. **Team member contribution:** This subsection should clearly identify the specific tasks, modules, and responsibilities completed by each team member throughout the project.

The description should be sufficiently detailed to allow the instructor to understand the contribution of each student to the project's design, implementation, testing, documentation, and presentation. All team members are expected to participate actively in the project, and individual contributions may be considered during the evaluation process.

8. **Conclusion:** The conclusion should summarize the work completed and the main achievements of the project. It should:

Review the project objectives and whether they were achieved.

Highlight the most important results and observations.

Discuss any limitations or challenges encountered.

Suggest possible improvements or future work.

The conclusion should not introduce new information, but rather provide a clear summary of the project's outcomes and lessons learned.

9. **References:** The references section should list all sources that were consulted or cited during the project. This includes:

Textbooks, research papers, and journal articles.

Online tutorials, documentation, and technical websites.

Software libraries, tools, and datasets used in the project.

Any AI tools (e.g., ChatGPT, Copilot, Gemini, Claude, etc.) that were used to assist in the project.

Guidelines:

Include only sources that were actually used.

Provide complete citation information so that the source can be easily located.

Use a consistent citation style throughout the report (e.g., IEEE, APA, or ACM).

All figures, tables, code snippets, and ideas obtained from external sources must be properly referenced.

Failure to acknowledge external sources may be considered plagiarism.

Note: The use of AI tools is permitted in this project; however, students must clearly disclose any AI tools used and briefly describe how they were used (e.g., code generation, debugging assistance, writing support, literature search, etc.).

IEEE conference paper template is found on the course page at Moodle (itc.birzeit.edu).

Project description:

he project aims to develop a machine-learning system that can distinguish between real human speech and AI-generated/deepfake speech. The main idea is to investigate whether traditional speech-processing features can capture acoustic artifacts introduced by modern speech synthesis and voice-conversion systems.

Suggested Datasets**1. ASVspoof 2021 – Speech Deepfake (DF) — Recommended**

The ASVspoof 2021 DF dataset contains bona fide human speech and spoofed speech generated using TTS and voice-conversion systems. It is specifically designed for speech deepfake detection and provides established evaluation protocols and baseline systems.

2. WaveFake

WaveFake contains 104,885 generated audio clips produced using several neural speech-generation architectures, including MelGAN, Parallel WaveGAN, Multi-Band MelGAN, HiFi-GAN, and WaveGlow. It is particularly useful for investigating whether detectors generalize across different synthesis methods.

The project can compare several traditional speech/signal-processing features:

Cepstral features

- MFCC — Mel-Frequency Cepstral Coefficients
- LFCC — Linear-Frequency Cepstral Coefficients
- CQCC — Constant-Q Cepstral Coefficients

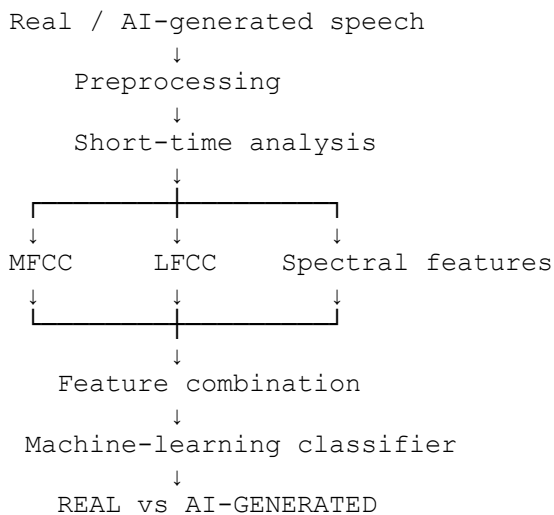
Spectral features

- Magnitude spectrum / FFT
- Mel-spectrogram
- Spectral centroid
- Spectral bandwidth
- Spectral roll-off
- Spectral flux

Time-domain features

- Short-time energy
- Zero-crossing rate

Suggested Experimental Design



A good first experiment would compare:

1. **MFCC + SVM**
2. **LFCC + SVM**
3. **Spectrogram + CNN**
4. **Combination of MFCC + spectral features**

The project could then investigate which features are most effective and whether a detector trained on one type of speech generator can detect speech produced by unseen generators. This last experiment is particularly valuable because WaveFake reports differences among generators, especially in higher-frequency characteristics.

For a manageable project, I would use **ASVspoof 2021 DF + MFCC/LFCC + SVM/Random Forest**, and then optionally add a **CNN using spectrograms** as an advanced comparison.

This gives a clear connection between traditional speech processing, machine learning, and the modern problem of AI-generated speech detection.

Final Project Deliverables by Every Group:

- Source code
- Final report
- Discussion/Presentation with Live demonstration

Evaluation Rubric:

Component	Marks	What is evaluated
1. Problem Understanding & Background	10	Understanding of AI-generated speech, TTS/voice conversion, and the deepfake detection problem
2. Dataset & Experimental Design	15	Appropriate dataset selection, preprocessing, train/validation/test split, avoiding data leakage, and justification of choices
3. Speech/Spectral Feature Extraction	20	Correct implementation and understanding of MFCC, LFCC, spectrogram/spectral features, and/or other selected features
4. Machine Learning / Classification	15	Appropriate classifier selection, training procedure, hyperparameters, and implementation
5. Evaluation & Results	15	Accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC/EER, and meaningful interpretation
6. Comparative Analysis	10	Comparison between features and/or classifiers; explanation of why some approaches perform better
7. Generalization / Robustness	5	Testing on unseen speakers, synthesis methods, or recording conditions
8. Report Quality	5	Organization, technical writing, figures, tables, references, and reproducibility
9. Presentation & Demonstration	5	Clear presentation, system demonstration, and ability to explain the work
Total	100	